Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Recombinant Spider Silk Wound Dressing			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Functional Restoration/Assistive & Welfare Devices	Medical Materials	Functional Restoration/Assistive & Welfare Devices	Bio-Materials
Concept of Proposed Technology	**1. Project Overview** The objective is to develop a sheet-type wound dressing utilizing a **non-animal-derived recombinant spider silk protein**. This biomaterial is engineered to address the mechanical and biological requirements of chronic wound management. The technology utilizes a recombinant production platform to ensure a consistent, scalable supply chain without the regulatory or safety concerns associated with animal-derived materials. At this stage, the project focuses on material characterization, pilot device evaluation, and generation of regulatory-relevant preclinical evidence, rather than product approval or commercialization. **2. Technical and Material Requirements** Chronic wound care necessitates dressings that provide structural integrity, flexibility, and permeability under moist physiological conditions. This proposed technology addresses these requirements through the following material properties: • **Mechanical Integrity:** The protein architecture consists of glycine- and alanine-rich sequences forming crystalline beta-sheet domains. This molecular structure provides the tensile strength and elasticity required to conform to irregular wound beds without mechanical failure.			

	• **Permeability and Barrier Function:** The matrix acts as a semi-permeable barrier, facilitating oxygen and water vapor exchange while providing a physical defense against exogenous bacterial colonization. • **Hemostatic Properties:** The material provides a scaffold for platelet and fibrin deposition, assisting in initial wound stabilization. **3. Biological Performance and Biocompatibility** The recombinant silk protein is designed for high biocompatibility and low immunogenicity to minimize inflammatory responses. Performance metrics include: • **Cellular Interaction:** Inclusion of specific spidroin motifs (e.g., GGX sequences) facilitates integrin-mediated cell adhesion. This supports the migration of keratinocytes and fibroblasts, as well as angiogenesis and collagen deposition. • **Biodegradation:** The material undergoes gradual macrophage-mediated degradation into non-toxic byproducts, reducing the frequency of dressing changes and minimizing secondary tissue trauma. • **Healing Cascade Support:** The material supports re-epithelialization through structural and biochemical cues rather than pharmacological action.
Rationale for Joint R&D Project	**1. Statement of Need** Chronic wounds—specifically diabetic foot ulcers, pressure ulcers, and venous leg ulcers—represent a persistent challenge in clinical wound management. Current dressing technologies often demonstrate sub-optimal efficacy in facilitating closure within complex, non-healing wound environments. There is a documented requirement for advanced solutions that provide structural support and a biocompatible microenvironment through material-driven mechanisms rather than pharmacological or biological interventions. **2. Technical Objectives and Scope** The proposed project involves the development of a wound dressing utilizing a **recombinant spider silk biomaterial**. As this material has no prior FDA registration history, the scope of work includes: • **Material Characterization:** Comprehensive analysis of physicochemical properties to establish a baseline for regulatory review. • **Design Optimization:** Engineering the device to meet specific clinical performance standards for chronic wound care. • **GMP-Compliant Manufacturing Preparation**: Production of recombinant spider silk protein and pilot devices at GMP-compliant facilities • **Validation and Translation:** Rigorous performance testing to support a 510(k) or De Novo regulatory pathway as a medical device. • **Regulatory Strategy and Documentation**: FDA regulatory strategy development, including pathway assessment, documentation planning, and preparation of technical and regulatory dossiers aligned with FDA expectations. **3. Implementation Framework (Joint R&D)** The project utilizes a collaborative model to integrate engineering manufacturing with *in vitro* and *in vivo* validation: Lead Organization (Company) • Engineering of recombinant biomaterials and scalable manufacturing. • Device design, prototyping, and technical performance validation. • Manufacturing and quality managing the medical device in accordance with GMP. • Execution of *in vivo* safety and biocompatibility studies in accordance with FDA/ISO standards. Collaboration Partner (Johns Hopkins University) • *In vitro* bench testing for material and device characterization. • *In vitro* and/or *in vivo* efficacy evaluation (POC level). • Scientific advisory input on *in vivo* safety study design and interpretation of results.

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	• Translational research support aligned with US regulatory expectations **4. Regulatory and Translational Strategy** The development program is intentionally categorized under a **medical device regulatory framework** rather than a drug or biologic pathway. This strategy is designed to: 1. Ensure broad clinical utility across diverse wound types. 2. Streamline the translational timeline from development to market. 3. Focus on material-based structural and biological functionality to minimize regulatory complexity.
Key Words	1. Recombinant Spider Silk Protein 2. pressure ulcer 3. Chronic Wound Dressing 4. Medical Device 5. Diabetic foot ulcer

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (6)
R&D Objectives	The project aims to achieve the following objectives: 1. Design and development of a chronic wound dressing incorporating a novel recombinant spider silk protein. 2. Quantitative evaluation and optimization of material and device performance, including mechanical, physical, functional properties, and biocompatibility tests relevant to wound dressing use. 3. Maintain flexibility between a 510(k) pathway and a De Novo pathway depending on regulatory feedback.			
R&D Target Specifications	The following target specifications are proposed to guide the R&D activities and may be refined during the project: • **Recombinant protein production yield**: ≥ 10 g/L (fermentation-based production, pilot-scale) • **Device form factor**: Sheet-type wound dressing (primary), with expandable design considerations for alternative formats • **Mechanical and physical properties**: ○ Tensile strength and/or flexibility suitable for chronic wound coverage ○ Structural integrity under moist wound conditions ○ Handling stability during clinical application • **Biological Characterization and non-clinical efficacy:** ○ *In-vitro* evaluation of cytotoxicity, biodegradability, and antimicrobial properties ○ *In-vitro* analysis of cell adhesion, proliferation, and regenerative potential (Investigation of cellular mechanisms underlying enhanced focal cell adhesion and subsequent cell regeneration) ○ *In-vivo* confirmation of biocompatibility and safety ○ *In-vivo* validation of tissue regeneration and wound healing efficacy observed *in vitro* • **Biocompatibility**: Compliance with ISO 10993 requirements for medical devices intended for skin and wound contact			

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	• **Regulatory readiness**: Generation of technical data supporting a medical device–based regulatory pathway (FDA), including preparation for Q-submission and IDE design, if required
R&D Plan	**Phase 1: Technology Development and Feasibility** (Years 1~2) This phase focuses on early technical feasibility, material/device characterization, and regulatory preparedness. Key activities include: • Optimization of sheet-type wound dressing design using the recombinant biomaterial • Quantitative material and device performance testing, including mechanical properties, stability, and functional evaluation • Biocompatibility assessments in accordance with relevant international standards • Preparation for regulatory interactions, including FDA Q-submission and preliminary IDE design, if applicable **Phase 2: Translational Development and Regulatory Advancement** (Years 2~3) This phase supports deeper translational preparation and regulatory pathway readiness. Key activities include: • Generation of efficacy-oriented data to support regulatory decision-making • Usability-informed design refinement • GMP-aligned documentation and risk management preparation • Preparation of FDA submission materials (510(k) or De Novo, as appropriate)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Research institute will contribute primarily to the clinical and translational aspects of the project, leveraging its expertise in biomedical research rather than serving as a clinical service provider. Expected roles and responsibilities include: • ***In vitro* bench testing for characterization** of the recombinant spider silk material and sheet-type pilot devices (e.g., cytocompatibility, cell adhesion, migration-related assays). • **Exploratory *in vitro* and/or non-GLP *in vivo* efficacy studies**, as appropriate, to assess biological performance relevant to chronic wound environments. • **Evaluation of pilot device compatibility with intended use**, focusing on material–tissue interaction, handling properties, and functional suitability. This evaluation can be conducted as a part of *in vivo* efficacy study. • **Scientific input on *in vivo* safety study design**, including study endpoints and interpretation frameworks (execution led by the company via qualified testing facilities). • **Preclinical translational guidance** to ensure that generated data are suitable for FDA regulatory pathways (510(k), and De Novo if required). • **Advisory support for regulatory documentation readiness**, including scientific input for FDA Q-submission and potential IDE-enabling data packages.
Expected R&R of Korean SME Partner	Korean SME partner is expected to serve as the **technology developer and industrial partner**, responsible for biomaterial engineering, device development, and manufacturing-oriented activities. Expected roles and responsibilities include: • **Development and optimization of recombinant spider silk protein**, including production process control, reproducibility, and scalability. • **Manufacturing of pilot-scale materials and devices** at GMP- or GMP-compatible facilities through qualified OEM partners. • **Design, fabrication, and iteration of sheet-type wound dressing** suitable for chronic wound applications. • **Execution of quantitative material and device performance evaluations**, including mechanical, physical, and functional testing. • **Conduct of biocompatibility and safety assessments** in accordance with applicable international standards (eg. ISO 0993). • **Preparation of technical documentation and data packages** supporting a medical device–

	based regulatory pathway. • **Overall project management**, including integration of R&D, manufacturing, regulatory, and translational activities.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed project is expected to generate the following outcomes and impacts: **1) Technological Outcomes** • Establishment of a novel recombinant spider silk–inspired biomaterial platform suitable for medical device applications, with no prior FDA registration precedent • Development and validation of a sheet-type wound dressing prototype optimized for chronic wound environments • Generation of quantitative performance data covering material properties, device functionality, and biocompatibility, supporting translational development • Accumulation of regulatory-ready technical data to support a medical device–based pathway, including preparation for FDA Q-submission and IDE-related activities, if applicable **2) Clinical and Translational Impact** • Identification of clinically relevant indications and application scenarios for chronic wounds, including diabetic foot ulcers, pressure ulcers, and venous leg ulcers • Early validation of clinical feasibility and usability of the proposed wound dressing through structured translational research • Establishment of a clear and efficient pathway toward clinical adoption, enabled by a device-focused development strategy rather than drug or biologic pathways • Provision of a practical technological alternative addressing unmet medical needs in chronic wound care, where existing wound dressing solutions demonstrate limited effectiveness **3) Commercialization Potential** • Demonstration of commercial feasibility through scalable recombinant protein production and device-oriented product design suitable for industrial manufacturing • Establishment of a solid foundation for future product commercialization, including preparation for global regulatory submissions and subsequent global market entry • Strengthening the position of the Korean SME partner as a global, technology-driven company capable of translating advanced biomaterials into internationally competitive medical device products **4) Industrial and Global Impact** • Contribution to the global competitiveness of the Korean medical device industry by introducing a first-in-class biomaterial-based wound care technology • Promotion of international collaboration between Korean SMEs and leading overseas research institutions through the TIPA framework • Expansion of opportunities for future global partnerships, licensing, and follow-on joint R&D projects beyond the initial scope of this proposal • Transfer and integration of global-leading clinical experience and perspectives on unmet clinical needs in the United States into Korean SME–driven medical device development Overall, this project is designed as a regulatory-oriented, preclinical joint R&D program that bridges advanced biomaterial innovation with practical medical device development. By combining the industrial and manufacturing leadership of a Korean SME with the translational research strengths of a leading U.S. research institution, the project aims to de-risk the development of a challenging first-in-class biomaterial and establish a credible pathway toward FDA approval and global market entry.

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6. References

(1) Chung, H., Kim, J., Lee, Y. J., Choi, K. R., Jeong, K. J., Kim, G.-J., & Lee, S. Y. (2023). Enhanced production of difficult-to-express proteins through knocking down rnpA gene expression. *Biotechnology Journal*, *18*(10), e2200641. https://doi.org/10.1002/biot.202200641

(2) Holland, C., Numata, K., Rnjak-Kovacina, J., & Seib, F. P. (2019). The Biomedical Use of Silk: Past, Present, Future. In Advanced Healthcare Materials (Vol. 8, Issue 1). Wiley-VCH Verlag. https://doi.org/10.1002/adhm.201800465

(3) Liebsch, C., Bucan, V., Menger, B., Köhne, F., Waldmann, K. H., Vaslaitis, D., Vogt, P. M., Strauss, S., & Kuhbier, J. W. (2018). Preliminary investigations of spider silk in wounds in vivo — Implications for an innovative wound dressing. Burns, 44(7), 1829–1838. https://doi.org/10.1016/j.burns.2018.03.016

(4) Rising, A., Widhe, M., Johansson, J., & Hedhammar, M. (2011). Spider silk proteins: Recent advances in recombinant production, structure-function relationships and biomedical applications. In Cellular and Molecular Life Sciences (Vol. 68, Issue 2, pp. 169–184). https://doi.org/10.1007/s00018-010-0462-z

(5) Salehi, S., Koeck, K., & Scheibel, T. (2020). Spider silk for tissue engineering applications. In Molecules (Vol. 25, Issue 3). MDPI AG. https://doi.org/10.3390/molecules25030737

(6) Xia, X.-X., Qian, Z.-G., Ki, C. S., Park, Y. H., Kaplan, D. L., & Lee, S. Y. (2010). Native-sized recombinant spider silk protein produced in metabolically engineered Escherichia coli results in a strong fiber. *Proceedings of the National Academy of Sciences of the United States of America*, *107*(32), 14059–14063. https://doi.org/10.1073/pnas.1003366107

(7) Zhao, L., Chen, D., Yao, Q., & Li, M. (2017). Studies on the use of recombinant spider silk protein/polyvinyl alcohol electrospinning membrane as wound dressing. International Journal of Nanomedicine, 12, 8103–8114. https://doi.org/10.2147/IJN.S47256